

RIDO Green Corridor and EV Battery Thermal Protocol

1. Extreme Ambient Temperature Protocol: When ambient temperature exceeds 40C in summer corridors, EV battery thermal management system BTMS must maintain internal cell temperature between 25C and 35C. If battery temperature exceeds 48C, trigger mandatory thermal throttling with speed capped to 50 km/h.

2. Dynamic Charging Corridor Selection: Long-haul EV trips >300 km must schedule high-power 120kW DC charging stops only when State-of-Charge reaches 15% to 25% window to maximize charging curve efficiency. Never fast-charge past 85% at public commercial hubs.

3. Regenerative Braking and Payload Compensation: For heavy commercial EVs descending ghat / hilly terrain gradient > 4%, drivers must engage Stage-3 Regenerative Braking to recover up to 18% kinetic energy into the traction battery while avoiding mechanical brake fade.